

Supplementary Results

Blind Color Image Watermarking Based on Reduced Biquaternion SVD and Quantization Index Modulation

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S1 Attack-wise operating-point maps

This section gives the attack-wise results underlying the operating-point analysis in Section 4.2 of the main article. The block size and QIM step are varied over $d \in \{4, 6, 8, 10, 12, 16\}$ and $\Delta \in \{40, 60, 80, 100, 120, 140, 160\}$. All cells are evaluated at $r = 1$, $K_{\max} = 2$, $\tau = \Delta/8$, and block-selection seed $\kappa = 2026$, with order clipping and adjacent-target fallback enabled. Each cell reports the bit error rate (BER) averaged with equal weight over the 32 host–watermark pairs.

Figures S1–S5 give the 19 condition-wise maps, including the no-attack control. For the Gaussian-noise and salt-and-pepper attacks, each cell is the equal-weight mean over the 32 pairs and ten seeds, $7, \dots, 16$; the remaining conditions are deterministic. Because each panel uses its own color scale, comparisons between panels should be based on the annotated cell values.

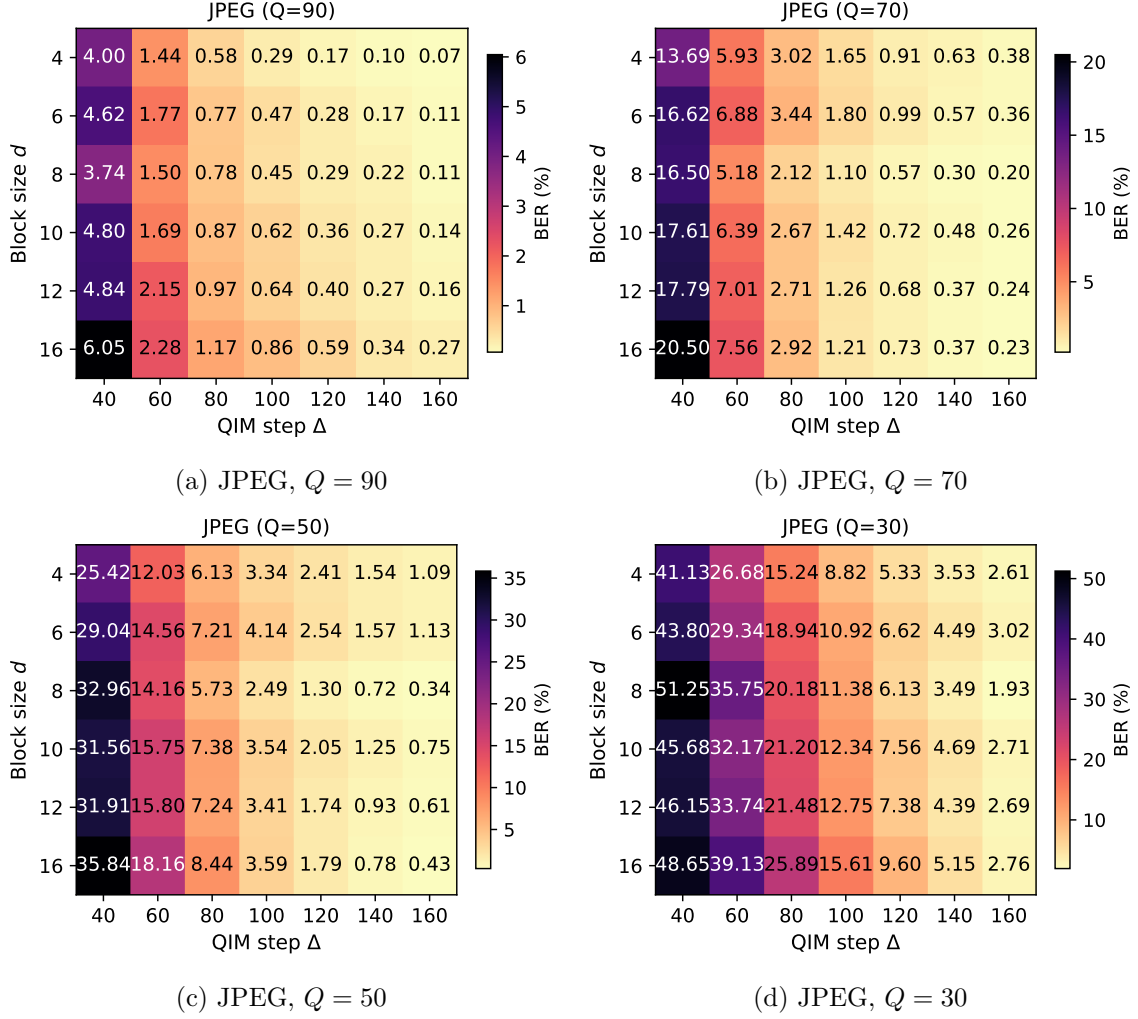


Figure S1: Pair-equal BER (%) over the 32 host–watermark pairs for JPEG compression at (a) $Q = 90$, (b) $Q = 70$, (c) $Q = 50$, and (d) $Q = 30$. Each cell gives the deterministic mean for one tested (d, Δ) combination. For every fixed d and Q , BER decreases monotonically as Δ increases; in general, it increases as Q is reduced from 90 to 30. The grid minima for all four JPEG levels occur in the $\Delta = 160$ column.

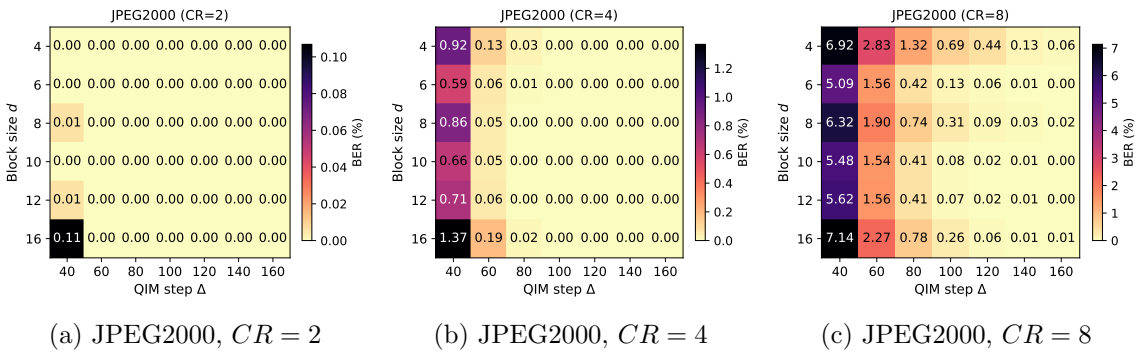


Figure S2: Pair-equal BER (%) over the 32 host–watermark pairs for JPEG2000 compression ratios (a) $CR = 2$, (b) $CR = 4$, and (c) $CR = 8$. Each cell gives the deterministic mean for one tested (d, Δ) combination. For every d , BER is nonincreasing with Δ at all three compression ratios. At $CR = 2$, BER is zero in 39 of the 42 cells and does not exceed 0.107%. At $CR = 4$, it is zero for every d at $\Delta = 100, 120, 140$, and 160. At $CR = 8$, the minimum is zero at $(10, 160)$ and $(12, 160)$.

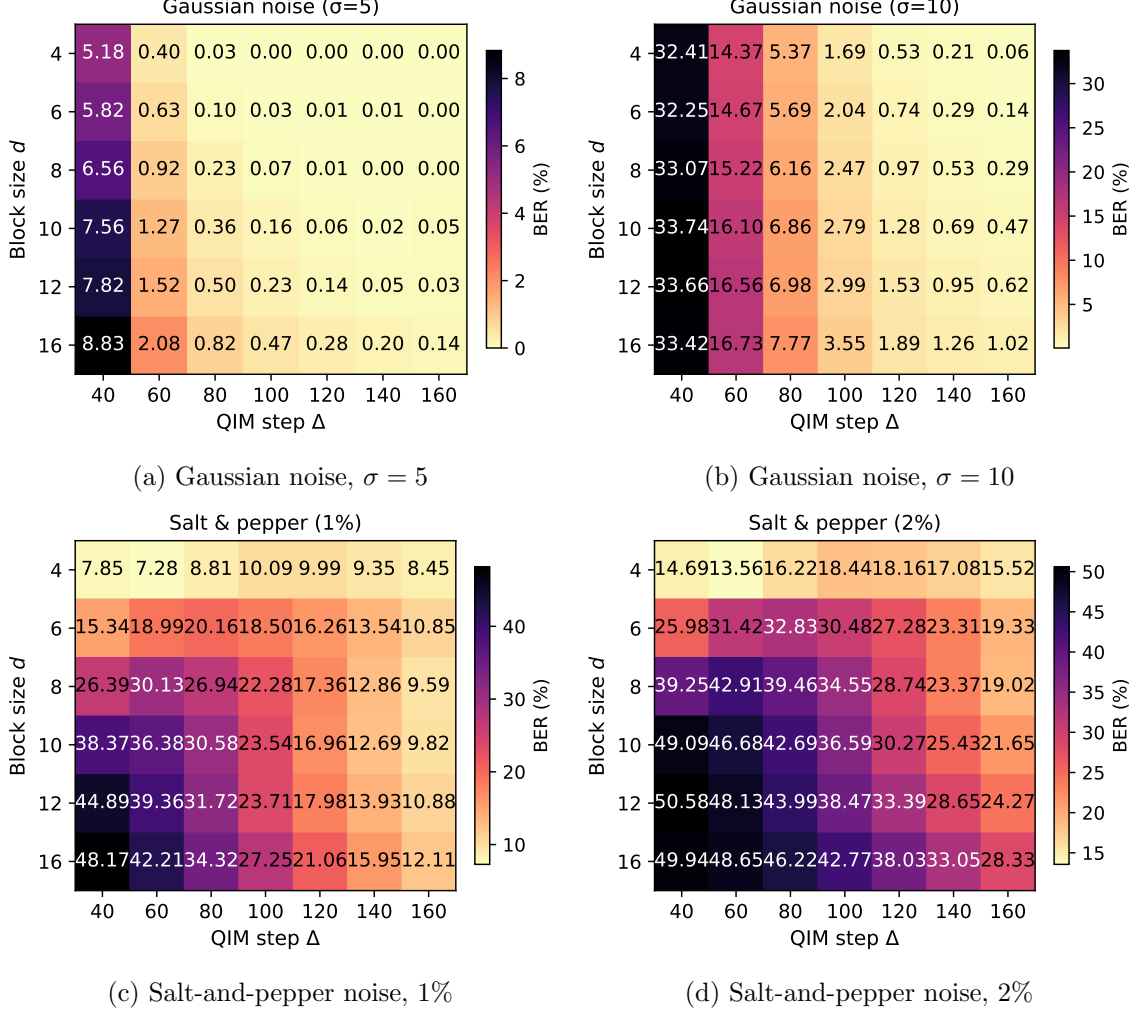


Figure S3: BER (%) for (a) Gaussian noise with $\sigma = 5$, (b) Gaussian noise with $\sigma = 10$, (c) 1% salt-and-pepper noise, and (d) 2% salt-and-pepper noise. Each cell is the equal-weight mean over the 32 host–watermark pairs and ten seeds, $7, \dots, 16$. The Gaussian-noise response generally improves as Δ increases; at $\sigma = 5$, BER is zero at $(4, 160)$, $(6, 160)$, and $(8, 160)$, while the $\sigma = 10$ minimum occurs at $(4, 160)$. For both salt-and-pepper levels, the grid minimum occurs at $(4, 60)$, and the response is not monotone in Δ .

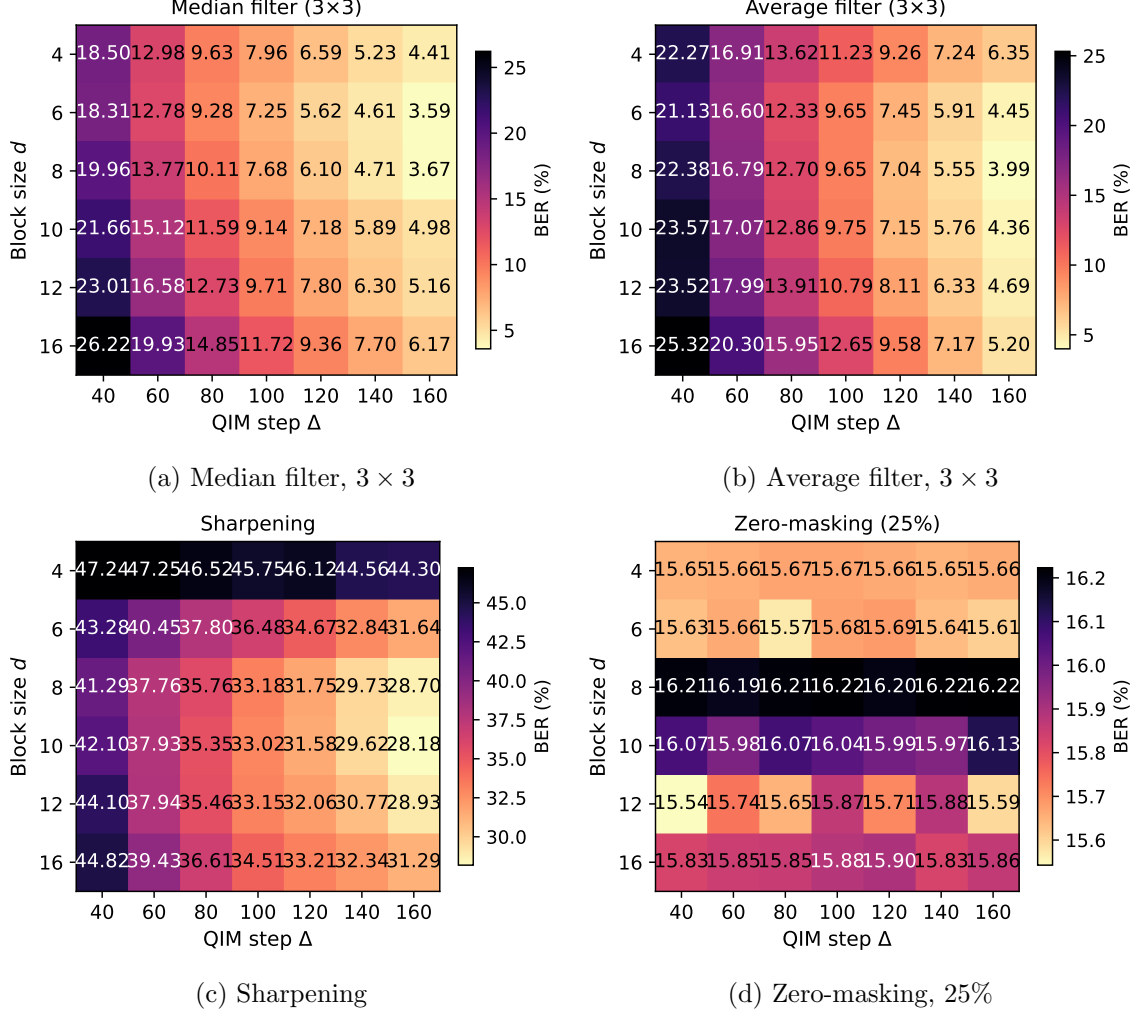


Figure S4: Pair-equal BER (%) over the 32 host-watermark pairs for (a) 3×3 median filtering, (b) 3×3 average filtering, (c) sharpening, and (d) 25% zero-masking. Each cell gives the deterministic mean for one tested (d, Δ) combination. Median- and average-filter BER decrease monotonically with Δ for every d , and their grid minima occur at $(6, 160)$ and $(8, 160)$, respectively. Sharpening reaches its minimum at $(10, 160)$. Zero-masking has no monotone response to Δ and remains between 15.543% and 16.223% throughout the grid.

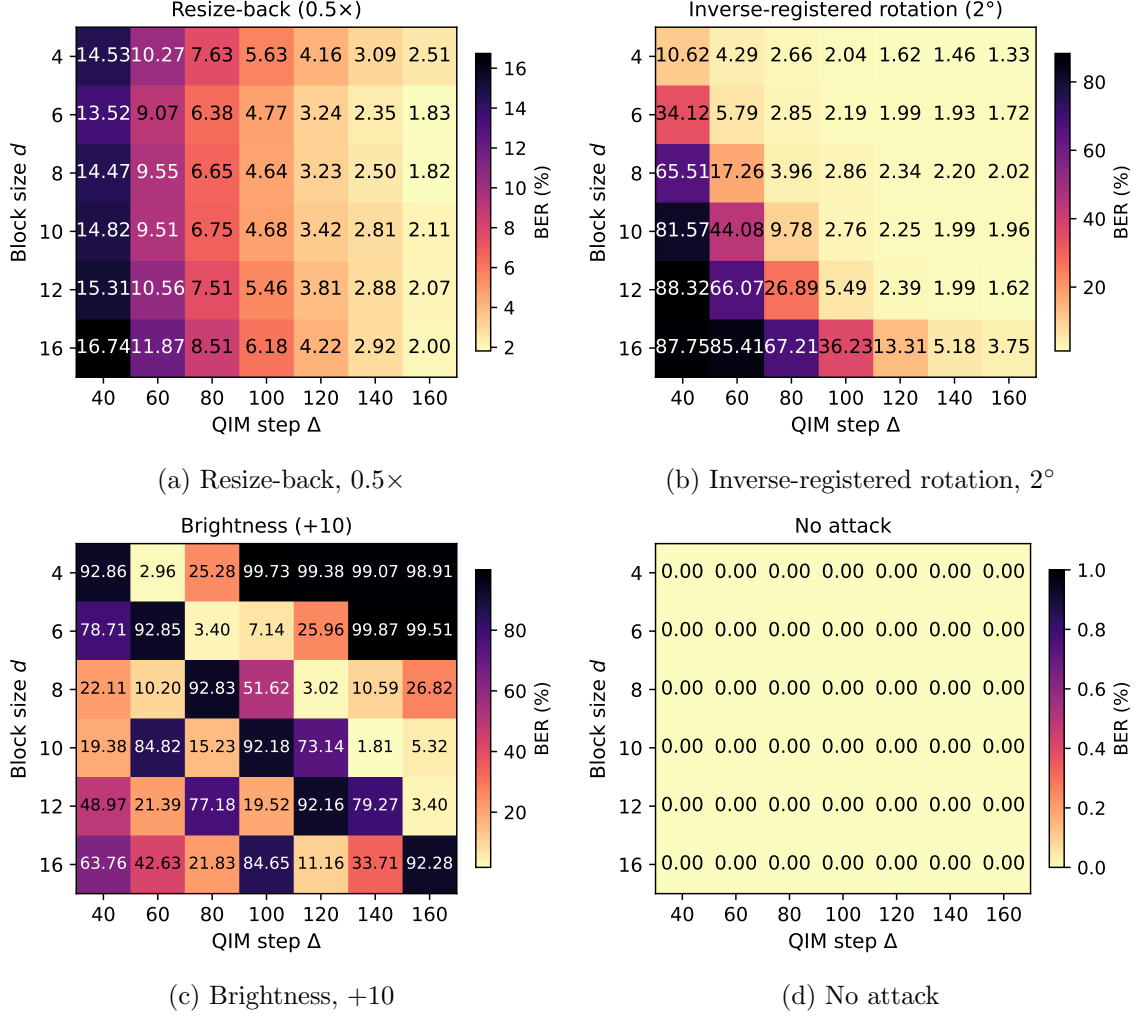


Figure S5: Pair-equal BER (%) over the 32 host–watermark pairs for (a) 0.5 $\times$ resize-back, (b) 2 $^\circ$ inverse-registered rotation, (c) brightness +10, and (d) the no-attack control. Each cell gives the deterministic mean for one tested (d, Δ) combination. For every d , BER under resize-back and inverse-registered rotation decreases monotonically with Δ ; their grid minima occur at (8, 160) and (4, 160), respectively. Brightness +10 is strongly nonmonotone, ranges from 1.813% to 99.875%, and reaches its grid minimum at (10, 140). The no-attack BER is zero in all 42 cells.

S2 Brightness-offset sensitivity

The brightness +10 condition reported in Section 4.3 of the main article is one sample from the full integer-offset sweep. At the selected operating point $(d, \Delta, r, K_{\max}) = (8, 140, 1, 2)$, each integer offset in $[-20, 20]$ is evaluated over the same 32 RBSVD-QIM host-watermark pairs, with zero serving as the no-attack reference.

Figure S6 gives the aggregate response and the variation among the 32 pairs. Figure S7 then separates the results by host image and watermark.

Define the residue

$$p_{\Delta}(z) = z - \Delta \lfloor z/\Delta \rfloor \in [0, \Delta).$$

The blind QIM decision rule used in the main article is

$$D_{\Delta}(z) = \begin{cases} 0, & p_{\Delta}(z) < \Delta/2, \\ 1, & p_{\Delta}(z) \geq \Delta/2. \end{cases} \quad (\text{S1})$$

Remark S1 (QIM-phase interpretation of the brightness sweep). For $w \in \{0, 1\}$, a QIM center has the form

$$q_w = k\Delta + \frac{\Delta}{4} + w\frac{\Delta}{2}, \quad k \in \mathbb{Z}.$$

If an attack changes the feature by x , then

$$\mathbf{1}\{D_{\Delta}(q_w + x) \neq w\} = \mathbf{1}\left\{p_{\Delta}(x) \in \left[\frac{\Delta}{4}, \frac{3\Delta}{4}\right)\right\}.$$

Thus, the ideal error indicator is independent of the embedded bit and has a square-wave form with period Δ .

To relate this response to a brightness offset, consider an ideal constant-grayscale $d \times d$ block with channel value a . After applying the same offset c to all three channels, assume $0 \leq a + c \leq 255$, so that no clipping occurs. Its component matrices are

$$A_1(c) = (a + c)(1 + 2i)J_d, \quad A_2(c) = -(a + c)J_d,$$

where J_d is the $d \times d$ all-ones matrix. Since J_d has the single nonzero singular value d , the paired $r = 1$ feature and its change are

$$\bar{\sigma}(c) = \frac{d(\sqrt{5} + 1)}{2}(a + c), \quad x(c) = \bar{\sigma}(c) - \bar{\sigma}(0) = \frac{d(\sqrt{5} + 1)}{2}c.$$

The corresponding nominal brightness period is therefore

$$T_c = \frac{2\Delta}{d(\sqrt{5} + 1)}.$$

For $d = 8$ and $\Delta = 140$, this gives $T_c \approx 10.82$, consistent with the repetition over roughly 10–11 offset units in Figure S6. Natural-image blocks need not exhibit this exact linear response; post-correction residuals and clipping introduce additional variation. Averaging these shifted block responses therefore smooths the ideal square wave.

The phase dependence makes a single brightness offset unsuitable for the Kodak24 published-method benchmark. For example, the mean BER is 99.878% at offset +7, 1.596% at +13, and 10.587% at +10; none of these values alone gives a representative summary. Brightness is therefore excluded from that benchmark and is instead reported through the full sweep in Figures S6 and S7.

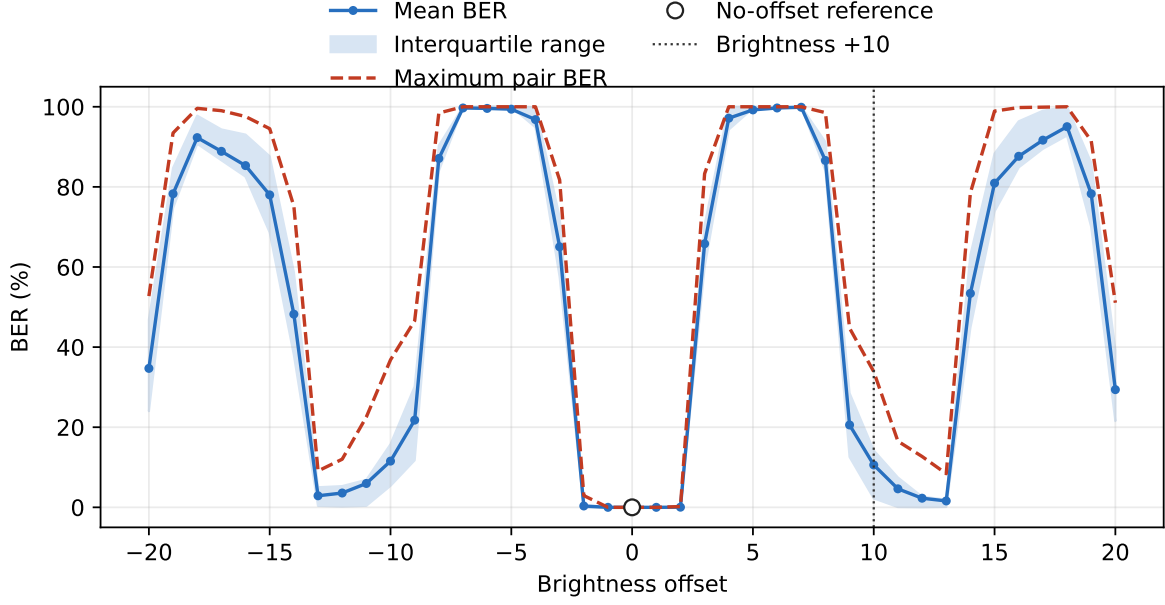


Figure S6: Uniform brightness-offset sweep for the 32 equally weighted host–watermark pairs at $(d, \Delta, r, K_{\max}) = (8, 140, 1, 2)$. The solid blue curve gives the pair-equal mean BER, the shaded band spans the 25th–75th percentiles, and the dashed red curve gives the maximum pair BER at each offset. The open circle at zero is the no-offset reference, and the vertical dotted line marks the brightness +10 condition reported in the controlled comparison. The mean BER ranges from 0% to 99.878% and exhibits an approximately periodic, square-wave-like response.

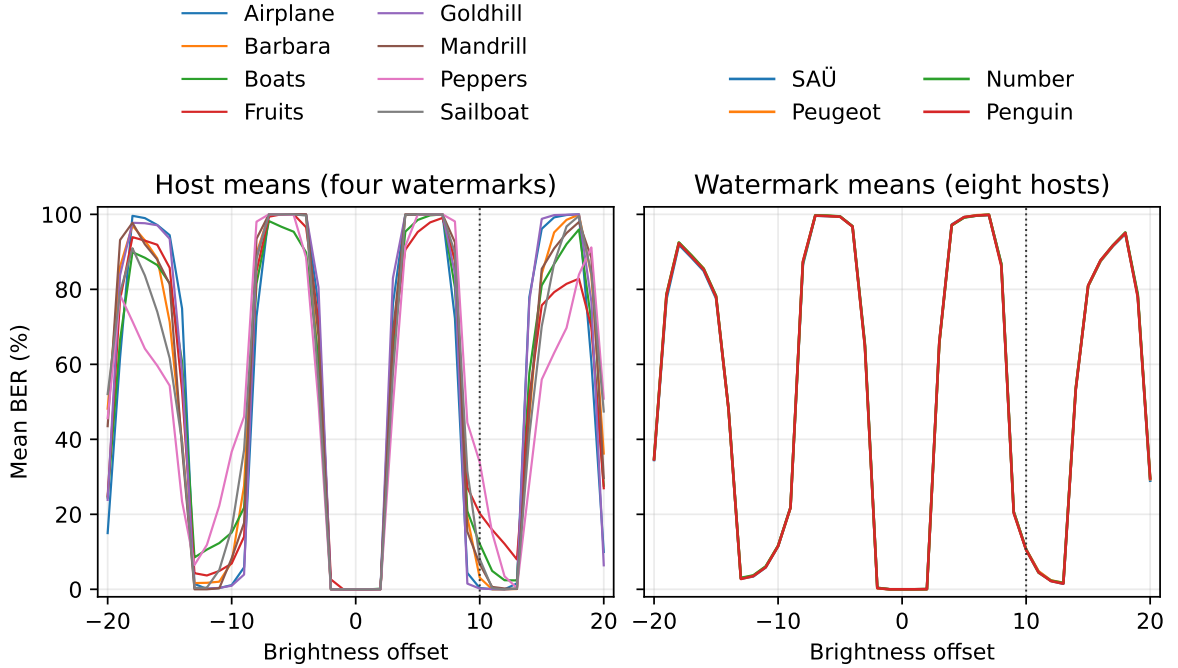


Figure S7: Brightness-offset response separated by host image and watermark. The left panel gives the mean BER of each host over the four watermarks, whereas the right panel gives the mean BER of each watermark over the eight hosts. The vertical dotted lines mark offset +10. At this offset, the host means range from 0.195% to 33.862%, while the watermark means range from 10.498% to 10.657%. Thus, the mean BER varies much more across hosts than across watermarks.

S3 Published-method implementation details

Table S1 records the implementation details of RBSVD–QIM and the nine published-method reimplementations used in the Kodak24 benchmark. It identifies the carrier or feature, processing region, selected setting, and adaptation required by the common four-watermark protocol. The image-quality and BER results are reported in Section 4.4 of the main article.

Table S1: Implementation details and common-protocol adaptations for RBSVD–QIM and nine reimplementations of published methods in the four-watermark Kodak24 benchmark. The proposed-method setting is the retained reference operating point used in the main article; each comparison strength is fixed across all 96 host–watermark pairs.

Method	Carrier or feature	Block or region	Selected setting	Common-protocol adaptation
Proposed RBSVD–QIM	Mean of the paired r -th singular values of the two reduced-biquaternion components	8×8 non-overlapping blocks	$(d, \Delta, r, K_{\max}) = (8, 140, 1, 2)$	No adaptation; evaluated directly at the retained reference operating point.
Sun et al. (2022) [1]	Largest singular value of the selected pure-quaternion block	2×2 quaternion blocks	$T = 128.950$	The common 1024-bit payload and PCG64 block order replace the published 24,576-bit RGB-watermark, Arnold-scrambling, and randperm arrangement; common inverse registration replaces the method-specific geometric-correction stage, and the nonnegative-target rule stated below is used.
Chen et al. (2021) [2]	Relative modulation of the i , j , and k components of q_{21} and q_{31} in the QQRD Q factor	4×4 quaternion blocks; up to three bits/block (342 blocks for 1024 bits)	$T_{\text{rel}} = 0.0623958$	The common payload and block order replace the Arnold/randperm payload arrangement.
Chen et al. (2023) [3]	Relative modulation of an adaptively selected imaginary-component pair of u_{21} and u_{31} in the QSVD U factor	4×4 quaternion blocks	$T = 0.0656149$	The common 1024-bit payload and PCG64 block order replace logistic-chaotic scrambling and RCS ordering; the conditional component-selection rule stated below is retained.
Chen et al. (2026) [4]	QIM of scalar entry $R_0(1, 4)$ in the QQR upper-triangular factor	4×4 quaternion blocks	$T = 0.555732$	The common 1024-bit payload and PCG64 block order are used.
Su et al. (2022) [5]	Block Frobenius norm defined by the printed Eq. (7)	4×4 scalar-channel blocks	$K = 2$; $QT_{\text{blue}} = 153.396$	A keyed 1024-bit subset of the eligible channel/block positions and the nonnegative-target rule stated below are used.
Li et al. (2020) [6]	QIM in the scalar part of the QQR R factor after QDFT, within the highest-entropy 128×128 block	Centered 512×512 region; selected 128×128 entropy block	$\phi = 106.646$	The common payload is mapped to the fixed upper-left 32×32 position set in the selected entropy block.
Hu et al. (2020) [7]	Mixed RM/QIM modulation of $u_{21} - u_{31}$ in the first left singular vector	8×8 scalar-channel SVD blocks	$\lambda = 216$; $D = 0.0603507$	A keyed 1024-position subset replaces full-capacity channel/block filling; the published embedding modules are retained.
WQSD [8]	Modulus $ t_{11} $ of the maximal-modulus quaternion-Schur diagonal entry	4×4 quaternion blocks	$T = 140.266$	The common payload and block order are used with the nonnegative-target rule stated below.
WQHD [8]	Magnitude $ h_{21} $ of the first quaternion-Hessenberg subdiagonal entry, implemented by the equivalent first-column tail norm	4×4 quaternion blocks	$T = 142.329$	The common payload and block order are used with the nonnegative-target rule stated below.

Calibration and code provenance. The RBSVD setting is the retained reference operating point defined in Section 4.2 of the main article. For each comparison method, several strength values are evaluated over all 96 pairs, and the value whose pair-equal mean PSNR is closest to the RBSVD mean is retained. The same value is then used for every pair; BER and SSIM are not used in this selection. The displayed strengths are rounded, whereas the implementations retain the full values used in the reported benchmark. Since these parameters act on different carriers, their numerical values are not directly comparable. The Supplementary Material contains all ten implementations used in the benchmark. No official implementation accompanied the source material examined; the nine comparison implementations were prepared for this study from the cited algorithms and were not released by the original authors.

Shared feasibility rules and source ambiguities. The Sun22, Su22, WQSD, and WQHD carriers are nonnegative. If the nearest same-bit QIM representative is negative, it is shifted forward by one QIM period to the first nonnegative representative on the same-bit lattice; the bit label is unchanged. Across the 98,304 embedded carriers per method at the reported strengths, this rule is triggered 4 times for Sun22, 261 times for Su22, and 78 times for WQHD, and is not triggered for WQSD. The Su22 implementation uses the Frobenius-norm expression printed in Eq. (7), although the paper describes it as an MSV. Its red- and green-channel steps are $QT_R = 0.78QT_{\text{blue}}$ and $QT_G = 0.94QT_{\text{blue}}$, respectively. The sentence following Eq. (14), which requires $\lambda_1^* \geq \lambda_1$ by adding QT_p when necessary, conflicts with the example in Figure 1, where 896.2539 is changed to 885.5000. The implementation therefore follows Eqs. (12)–(14) and Figure 1 and selects the nearer representative without this subsequent one-sided update.

Method-specific implementation choices. The Chen21 and Chen26 reimplementations share a direct quaternion-Givens QQR kernel. For Chen21, this replaces the paper’s real-representation, structure-preserving Householder kernel; it reproduces the printed QQR example to four decimal places but does not implement the published runtime optimization. For Chen26, the reimplementation preserves the published quaternion-Givens QQR and $R_0(1, 4)$ QIM carrier but does not implement the paper’s fast real-representation kernel.

The Chen23 paper gives no scalar rule for selecting the i, j, k embedding pair. For the benchmark, the pairs are ranked by $d_\ell = ||u_{21}^{(\ell)} - u_{31}^{(\ell)}| - T|$. Embedding ties are resolved by the smaller maximum coefficient change and then by the fixed i, j, k order. A selected pair that already satisfies the required bit inequality is left unchanged; otherwise, Eqs. (6)–(7) are applied. Extraction uses the same distance defined in Eq. (8), resolves ties in i, j, k order, and assigns a zero difference to bit 0.

For Li20, the unspecified entropy calculation and high-entropy choice are implemented as 256-bin, per-block min–max Shannon entropy followed by selection of the maximum. The common 1024-bit payload is mapped to the upper-left 32×32 positions of that block. The source-prescribed block search is repeated during extraction, so the selected block can change and contribute to no-attack decoding errors. A centered 512×512 region is processed, with all other pixels retained unchanged. For Hu20, the reimplementation uses deterministic bounded zero-mean dither, modified Gram–Schmidt restoration, and a 20-pass safety cap for the source-prescribed iterative regulation.

S4 Sensitivity results at $\Delta = 120$ and $\Delta = 160$

The main comparisons use the retained reference setting $(d, \Delta, r, K_{\max}) = (8, 140, 1, 2)$. At fixed $d = 8$, $r = 1$, and $K_{\max} = 2$, the auxiliary $\Delta = 120$ setting favors image quality, whereas $\Delta = 160$ favors robustness. This section reports the corresponding controlled and Kodak24 results for both settings.

S4.1 Controlled comparison

The auxiliary runs retain the controlled protocol of Section 4.3 of the main article. The eight host images and four 1024-bit watermarks give 32 host–watermark pairs, to which all reported pair-level summaries assign equal weight. All three methods use $d = 8$, $r = 1$, and $K_{\max} = 2$, with the same payload streams, keyed block positions, matched attack realizations, and order-preserving output-pixel correction procedure; adjacent-target fallback remains specific to RBSVD–QIM. For each reference step $\Delta_{\text{ref}} \in \{120, 160\}$, RBSVD–QIM and dense-adjoint QSVD–QIM use Δ_{ref} , whereas Channel-SVD–QIM uses the bound-matched step $\Delta_{\text{ref}}/\sqrt{3}$, namely 69.282 or 92.376. Deterministic conditions are evaluated once, and the Gaussian-noise and salt-and-pepper conditions use ten matched attack seeds, $7, \dots, 16$. Family-balanced BER is the equal average of the five within-family means; the no-attack and brightness conditions do not enter this score.

Table S2: Imperceptibility and aggregate robustness at the two auxiliary reference steps. PSNR and SSIM are pair-equal means $\pm$ descriptive sample standard deviations across the 32 host–watermark pairs. Within each reference-step group, boldface marks the largest mean PSNR and SSIM and the smallest family-balanced BER, determined from the unrounded values.

Method	Step used	PSNR mean±SD (dB)	SSIM mean±SD	Family-balanced BER (%)
(a) $\Delta_{\text{ref}} = 120$				
RBSVD–QIM	120	47.134 ± 1.069	0.996144 ± 0.001537	9.391
Channel-SVD–QIM	69.282	46.913 ± 1.005	0.995893 ± 0.001679	10.881
Dense-adjoint QSVD–QIM	120	46.921 ± 0.998	0.995961 ± 0.001678	10.121
(b) $\Delta_{\text{ref}} = 160$				
RBSVD–QIM	160	44.648 ± 1.032	0.993402 ± 0.002558	6.725
Channel-SVD–QIM	92.376	44.453 ± 1.006	0.992960 ± 0.002864	8.086
Dense-adjoint QSVD–QIM	160	44.441 ± 1.011	0.993086 ± 0.002843	7.270

Table S3 gives the corresponding condition-wise BER values.

Table S3: Attack-wise pair-equal BER (%) at the two auxiliary reference steps. Gaussian-noise and salt-and-pepper entries are means $\pm$ descriptive sample standard deviations across the ten matched attack seeds; the remaining entries are deterministic. Boldface marks the minimum within each three-method reference-step group, including all ties, based on the unrounded mean BER values; it does not indicate statistical significance. The no-attack and brightness conditions are included for completeness but are excluded from the family-balanced BER in Table S2.

Attack	$\Delta_{\text{ref}} = 120$			$\Delta_{\text{ref}} = 160$		
	RBSVD-QIM	Channel-SVD-QIM	Dense-adjoint QSVD-QIM	RBSVD-QIM	Channel-SVD-QIM	Dense-adjoint QSVD-QIM
No attack	0.000	0.220	0.000	0.000	0.220	0.000
JPEG ($Q = 90$)	0.293	0.543	0.168	0.107	0.369	0.061
JPEG ($Q = 70$)	0.568	0.665	0.385	0.198	0.415	0.104
JPEG ($Q = 50$)	1.303	1.932	1.505	0.336	0.668	0.363
JPEG ($Q = 30$)	6.131	11.194	10.028	1.929	3.586	2.994
JPEG2000 ($CR = 2$)	0.000	0.226	0.000	0.000	0.226	0.006
JPEG2000 ($CR = 4$)	0.000	0.250	0.000	0.000	0.229	0.006
JPEG2000 ($CR = 8$)	0.089	0.336	0.070	0.021	0.262	0.024
Gaussian noise ($\sigma = 5$)	0.013 $\pm$ 0.011	0.413 $\pm$ 0.014	0.018 $\pm$ 0.009	0.000 $\pm$ 0.000	0.310 $\pm$ 0.011	0.022 $\pm$ 0.001
Gaussian noise ($\sigma = 10$)	0.969 $\pm$ 0.123	1.849 $\pm$ 0.203	1.136 $\pm$ 0.163	0.287 $\pm$ 0.025	0.828 $\pm$ 0.028	0.372 $\pm$ 0.037
Salt & pepper (1%)	17.363 $\pm$ 0.561	20.338 $\pm$ 0.723	19.384 $\pm$ 0.630	9.588 $\pm$ 0.311	12.511 $\pm$ 0.547	11.002 $\pm$ 0.437
Salt & pepper (2%)	28.738 $\pm$ 0.343	32.313 $\pm$ 0.580	31.308 $\pm$ 0.453	19.019 $\pm$ 0.545	22.925 $\pm$ 0.648	21.036 $\pm$ 0.617
Median filter (3×3)	6.097	7.217	6.274	3.671	4.788	4.074
Average filter (3×3)	7.037	8.072	7.159	3.995	4.684	4.120
Zero-masking (25%)	16.199	16.434	16.220	16.223	16.406	16.214
Resize-back ($0.5 \times$)	3.232	3.705	3.180	1.819	2.228	1.700
Inverse-registered rotation (2°)	2.338	2.634	2.576	2.017	2.267	2.112
Brightness (+10)	3.021	1.065	1.337	26.825	4.529	12.366
Sharpening	31.747	36.285	33.423	28.696	33.591	30.664

Together with the retained $\Delta = 140$ result, the controlled RBSVD-QIM values give the sequences $47.134 \rightarrow 45.832 \rightarrow 44.648$ dB for mean PSNR, $0.996144 \rightarrow 0.994846 \rightarrow 0.993402$ for mean SSIM, and $9.391\% \rightarrow 7.874\% \rightarrow 6.725\%$ for family-balanced BER as Δ increases from 120 to 140 and then to 160. Thus, among these three evaluated settings, $\Delta = 120$ is quality-oriented, $\Delta = 160$ is robustness-oriented, and $\Delta = 140$ is the intermediate compromise retained for the main article. The three measured points do not establish a unique or global optimum.

S4.2 Kodak24 published-method benchmark

The two auxiliary Kodak24 benchmarks use the same 24 host images and four 1024-bit watermarks, giving 96 host–watermark pairs for each method. Pair-level quality measures receive equal weight. The payload streams, attack realizations, and nine published-method reimplementations are the same as in Section 4.4 of the main article. At each reference step, RBSVD–QIM uses $(d, \Delta, r, K_{\max}) = (8, \Delta_{\text{ref}}, 1, 2)$. For every comparison method, one global strength shared by all 96 pairs is selected by minimizing the absolute difference between its pair-equal mean PSNR and the RBSVD mean PSNR at that reference step; BER and SSIM do not enter the selection. This calibration is performed independently at $\Delta_{\text{ref}} = 120$ and 160. The strength symbols and fixed method-specific settings are defined in Table S1; since they act on different features, their numerical values are not directly comparable.

Deterministic attack entries pool $96 \times 1024 = 98,304$ bit decisions per method and condition. The Gaussian-noise and salt-and-pepper entries are means over ten matched seeds, $7, \dots, 16$, with 98,304 decisions per method at each seed. Brightness is omitted because its BER depends on the selected offset; Section S2 gives the corresponding offset-sensitivity analysis.

Table S4: Global strength selection and image quality for the auxiliary $\Delta = 120$ Kodak24 benchmark. PSNR and SSIM are pair-equal means over the 96 host–watermark pairs. The RBSVD target is 47.982351 dB. The selected Hu20 strength gives 47.982370 dB, an absolute gap of 0.000019 dB. The selected Su22 setting gives 47.936288 dB, a gap of 0.046063 dB, whereas every other comparison method lies within 0.000132 dB of the target.

Method	Strength used	PSNR (dB)	SSIM
Proposed RBSVD–QIM	$\Delta = 120$	47.982	0.996483
Sun et al. (2022) [1]	$T = 110.211$	47.982	0.994297
Chen et al. (2021) [2]	$T_{\text{rel}} = 0.0506445$	47.982	0.995181
Chen et al. (2023) [3]	$T = 0.0544238$	47.982	0.994886
Chen et al. (2026) [4]	$T = 0.472783$	47.982	0.994237
Su et al. (2022) [5]	$QT_{\text{blue}} = 131.529$	47.936	0.995268
Li et al. (2020) [6]	$\phi = 90.3881$	47.982	0.994261
Hu et al. (2020) [7]	$D = 0.0505560$	47.982	0.994700
WQSD [8]	$T = 120.030$	47.982	0.994906
WQHD [8]	$T = 121.817$	47.982	0.994273

Table S5: Attack-wise BER (%) in the auxiliary $\Delta = 120$ Kodak24 benchmark. RBSVD-QIM uses $(d, \Delta, r, K_{\max}) = (8, 120, 1, 2)$, and the nine reimplementations use the global strengths in Table S4. Values are rounded to three decimal places. Boldface marks every minimum and tie in a row, determined from the unrounded BER values.

Attack	RBSVD	Sun22	Chen21	Chen23	Chen26	Su22	Li20	Hu20	WQSD	WQHD
No attack	0.000	0.000	0.000	0.236	0.608	0.865	7.982	0.000	0.247	0.520
JPEG ($Q = 90$)	0.174	0.026	11.365	18.170	1.134	3.872	7.723	15.999	0.548	1.112
JPEG ($Q = 70$)	0.521	0.889	18.491	31.442	3.840	15.330	30.484	25.869	0.933	5.301
JPEG ($Q = 50$)	0.984	5.542	26.479	39.838	9.919	21.389	42.697	34.579	1.642	12.470
JPEG ($Q = 30$)	3.729	18.785	35.294	44.249	19.981	26.825	45.873	39.641	4.777	23.187
JPEG2000 ($CR = 2$)	0.000	0.000	0.028	0.497	0.615	0.758	7.881	0.001	0.267	0.555
JPEG2000 ($CR = 4$)	0.000	0.000	0.046	0.815	0.617	0.741	7.907	0.001	0.266	0.558
JPEG2000 ($CR = 8$)	0.002	0.025	1.502	5.356	0.720	1.075	6.301	0.204	0.420	0.668
Gaussian noise ($\sigma = 5$)	0.187	0.000	3.180	5.753	0.797	0.837	4.551	0.012	0.544	0.659
Gaussian noise ($\sigma = 10$)	2.148	0.695	11.788	15.037	1.628	1.675	20.290	0.314	1.617	1.201
Salt & pepper (1%)	19.468	2.193	0.739	3.826	4.667	6.246	36.113	4.545	10.671	2.027
Salt & pepper (2%)	31.503	4.401	1.658	7.272	8.512	10.835	42.729	8.576	19.444	3.561
Median filter (3×3)	8.074	28.353	48.680	49.649	38.187	4.588	61.227	48.024	6.769	38.308
Average filter (3×3)	8.201	22.029	48.803	49.849	34.623	5.817	78.389	49.791	7.701	34.685
Zero-masking (25%)	15.503	15.894	10.767	17.242	10.149	16.610	14.207	8.459	9.941	10.092
Resize-back ($0.5\times$)	5.223	11.207	19.606	25.375	27.327	3.563	42.985	14.024	2.995	28.834
Inverse-registered rotation (2°)	3.724	1.905	3.481	7.968	2.256	2.386	5.278	1.623	1.990	1.656
Sharpening	29.784	52.509	1.297	46.578	44.227	48.115	58.219	77.531	48.713	44.947

Table S6: Global strength selection and image quality for the auxiliary $\Delta = 160$ Kodak24 benchmark. PSNR and SSIM are pair-equal means over the 96 host–watermark pairs. The RBSVD target is 45.527142 dB. The selected Hu20 strength gives 45.527101 dB, an absolute gap of 0.000042 dB. The selected Su22 setting gives 45.508260 dB, a gap of 0.018883 dB, whereas every other comparison method lies within 0.000329 dB of the target.

Method	Strength used	PSNR (dB)	SSIM
Proposed RBSVD–QIM	$\Delta = 160$	45.527	0.994105
Sun et al. (2022) [1]	$T = 144.587$	45.527	0.990851
Chen et al. (2021) [2]	$T_{\text{rel}} = 0.0733462$	45.527	0.991905
Chen et al. (2023) [3]	$T = 0.0760908$	45.527	0.991849
Chen et al. (2026) [4]	$T = 0.636448$	45.527	0.990704
Su et al. (2022) [5]	$QT_{\text{blue}} = 173.620$	45.508	0.992612
Li et al. (2020) [6]	$\phi = 121.9437$	45.527	0.990382
Hu et al. (2020) [7]	$D = 0.0696696$	45.527	0.991060
WQSD [8]	$T = 160.291$	45.527	0.991644
WQHD [8]	$T = 159.543$	45.527	0.990698

Table S7: Attack-wise BER (%) in the auxiliary $\Delta = 160$ Kodak24 benchmark. RBSVD–QIM uses $(d, \Delta, r, K_{\text{max}}) = (8, 160, 1, 2)$, and the nine reimplementations use the global strengths in Table S6. Values are rounded to three decimal places. Boldface marks every minimum and tie in a row, determined from the unrounded BER values.

Attack	RBSVD	Sun22	Chen21	Chen23	Chen26	Su22	Li20	Hu20	WQSD	WQHD
No attack	0.000	0.000	0.000	0.107	0.873	1.199	16.709	0.000	0.402	0.746
JPEG ($Q = 90$)	0.096	0.009	10.214	13.699	0.990	2.179	5.544	12.611	0.776	1.073
JPEG ($Q = 70$)	0.225	0.471	14.306	25.987	1.848	12.478	23.181	21.117	0.987	3.359
JPEG ($Q = 50$)	0.456	2.328	19.920	35.298	6.239	19.000	37.317	28.347	1.287	10.013
JPEG ($Q = 30$)	1.085	12.170	29.204	41.804	12.943	24.596	43.719	36.572	1.997	19.073
JPEG2000 ($CR = 2$)	0.000	0.000	0.007	0.202	0.874	1.039	12.594	0.000	0.459	0.762
JPEG2000 ($CR = 4$)	0.000	0.000	0.008	0.349	0.879	1.043	12.621	0.000	0.460	0.768
JPEG2000 ($CR = 8$)	0.000	0.000	0.684	3.051	0.922	0.695	11.102	0.033	0.587	0.842
Gaussian noise ($\sigma = 5$)	0.357	0.000	1.425	3.201	0.910	1.167	8.660	0.003	1.134	0.818
Gaussian noise ($\sigma = 10$)	0.990	0.036	6.356	8.894	1.125	1.317	15.821	0.061	1.390	0.925
Salt & pepper (1%)	12.203	2.442	0.736	3.590	4.727	6.058	32.438	2.582	9.919	2.391
Salt & pepper (2%)	22.545	4.917	1.652	6.921	8.474	10.147	40.481	5.121	18.031	4.095
Median filter (3×3)	5.541	29.952	48.833	49.647	37.668	3.402	65.669	48.514	4.688	40.332
Average filter (3×3)	5.501	20.507	48.552	49.913	32.070	4.252	74.984	49.901	5.294	35.576
Zero-masking (25%)	15.503	15.894	10.767	17.168	10.341	16.804	11.551	8.459	10.050	10.239
Resize-back ($0.5 \times$)	3.156	8.634	15.720	21.595	24.479	2.391	42.859	10.647	1.967	29.806
Inverse-registered rotation (2°)	2.921	1.708	2.055	6.023	1.939	2.547	6.937	1.194	1.736	1.582
Sharpening	26.733	51.850	0.672	47.158	46.475	47.981	52.310	80.101	47.717	45.192

For RBSVD–QIM, the $\Delta = 120 \rightarrow 140 \rightarrow 160$ sequence gives $47.982 \rightarrow 46.634 \rightarrow 45.527$ dB in mean PSNR, $0.996483 \rightarrow 0.995256 \rightarrow 0.994105$ in mean SSIM, and $10.186\% \rightarrow 8.749\% \rightarrow 7.620\%$ in family-balanced BER. Thus, within this method and the tested protocol, increasing Δ lowers aggregate BER while reducing image quality. The no-attack BER of RBSVD–QIM is 0.000% at both auxiliary settings; its largest listed attacked-condition BER changes from 31.503% under 2% salt-and-pepper noise at $\Delta = 120$ to 26.733% under sharpening at $\Delta = 160$. Since every comparison method is independently recalibrated to the corresponding RBSVD PSNR target, changes in attack-wise winners across the three settings cannot be attributed to Δ alone. The comparison therefore describes a within-method quality–robustness change rather than a fixed-quality cross-setting ranking.

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